

✓ Homework 1: Efficient Data Manipulation Using Numpy

After going through the lecture notebooks, work on the homework below.

Question

Implement a function that takes a 1D NumPy array of integers and returns an array containing elements that are multiples of 5 and leave a remainder of 1 when divided by 2.

Inputs / Outputs:

```
homework(a: np.ndarray) -> np.ndarray
```

Argument	Type	Description
a	np.ndarray	1D NumPy array of integers
Return	Type	Description
result	np.ndarray	Array containing elements that are both multiples of 5 and odd

Example:

```
a = np.array([1, 5, 10, 3, 4, 25, 30])
result = homework(a)
print(result)
```

```
[ 5 25]
```

Submission Guidelines:

1. Import libraries
2. Implement `homework` that returns elements which are multiples of 5 and odd
3. Define a 1D NumPy array
4. Check output with `print`

Note:

- Write your solution where `!!WRITE ME!!` appears in `homework`. (Remove `!!WRITE ME!!` before submission.)

- Submit the full `homework` function in the Omnicampus homework submission page and click [Python code submission].
- The return value must be a numeric array.
- Keep everything in a single function.

Deadline

Wed, Apr 29 (11:00 UTC)

✓ 1. Importing Libraries

Note: Do not include this section in your `homework()` function.

```
# 1. Import libraries
import numpy as np
```

```
# 2. Define a 1D NumPy array
a = np.array([1, 5, 10, 3, 4, 25, 30])
```

✓ 2. Solution

Implement the function below and submit it to Omnicampus.

```
# 3. Define the homework function
# Fill in !!WRITE ME!! (remember that only this homework function should be submitted)
def homework(a):
    !!WRITE ME!!
    return my_result
```

✓ 3. Test Your Solution

Run the test cell below to check that your `homework()` function does not contain import statements or data loading code before submitting.

Note: Do not modify the test code.

```
import ast
import inspect

def hw_public_tests():
    print("=== Public Tests ===")

    source = inspect.getsource(homework)
    tree = ast.parse(source)

    # Test 1: Disallow import statements
    print("Test 1: Check for import statements...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, (ast.Import, ast.ImportFrom)):
            print("NG")
            print("  homework() contains import statements. Please remove them.")
            return
    print("OK")

    print("=== All Public Tests Passed ===")

hw_public_tests()
print(homework(a))
```

Once all tests have passed, copy and paste your `homework()` function to Omnicampus and submit your solution. If you see a `1.0`, it means that your answer was correct.

Double-click (or enter) to edit